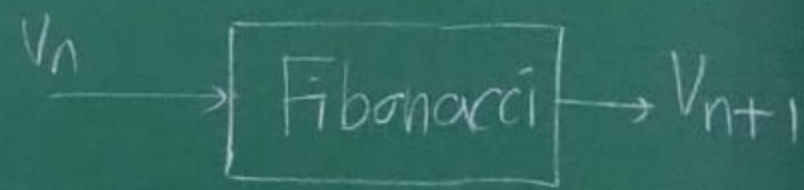


Fibonacci numbers $F_0 = 0$ $F_1 = 1$, $F_{n+2} = F_{n+1} + F_n$ for $n \geq 0$

Want: Formula for F_n



Think of this as a system evolving with time

State of system at stage n : " F_n, F_{n+1} " encoded as vector $\begin{bmatrix} F_{n+1} \\ F_n \end{bmatrix} = V_n$, So $V_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

Rule of evolution: $\begin{bmatrix} F_{n+2} \\ F_{n+1} \end{bmatrix} = \begin{bmatrix} F_{n+1} + F_n \\ F_{n+1} \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}}_A \begin{bmatrix} F_{n+1} \\ F_n \end{bmatrix}$ $F_n = \text{lower slot of } V_n$

Observe. $V_{n+1} = A V_n \Rightarrow V_n = A V_{n-1} = A^n V_0$

(Then $F_n = \text{bottom left entry of } A^n$)

Would like to be able to calculate arbitrary power of A easily

Fact! There is a matrix P such that $\bar{P}' A P = D = \begin{bmatrix} \alpha & 0 \\ 0 & \beta \end{bmatrix}$ (magic!)
(Can you find this "change of basis" matrix P and magic numbers α, β ?)

Upshot: $\bar{P}' A P = D \iff AP = PD \iff A = P D \bar{P}'$

$$A^n = P D^n \bar{P}' = P \begin{bmatrix} \alpha^n & 0 \\ 0 & \beta^n \end{bmatrix} \bar{P}'$$

F_n is of form (scalar) α^n + (scalar) β^n

Exercise Can you find a vector space and an operator on it to conceptualize what we did?

To be discussed
next week!

Regard an $n \times n$ matrix $A = \text{sequence } (C_1, \dots, C_n)$ of n col vectors in $\mathbb{R}^n$.

Thm There exists a unique multilinear alternating function

$D: (n \times n \text{ matrices}) \rightarrow \mathbb{R}$ such that $D(I) = u$, where u is a fixed real number

Remarks: • For $D = \det$, we take $u = 1$ • "Column" can be replaced by row, as we will see

• Defⁿ of "multilinear", "alternating": see below.

Properties ① $\det(A^T) = \det(A)$ ② $\det(AB) = \det(A) \det(B)$

② $\xrightarrow[\text{little work}]{\text{with a}}$ ③ $A \text{ is invertible} \iff \det(A) \text{ is invertible in } \mathbb{R} \text{ (i.e. nonzero)}$

④ $\det(A)$ can be calculated by row reduction | ⑤ $A \operatorname{adj}(A) = \det(A) I = \operatorname{adj}(A) A$

Defⁿ Let f be a function $\underbrace{V \times V \times \dots \times V}_{n \text{ times}} \rightarrow W$, where V, W are v-spaces
 We say that f is (For det, $V = \mathbb{R}^n$, $W = \mathbb{R}$)

(i) multilinear if f is linear in each of the n slots individually when other slots are fixed.
 e.g. linearity in 1st slot means $\begin{cases} f(v_1 + v'_1, v_2, \dots, v_n) = f(v_1, v_2, \dots, v_n) + f(v'_1, v_2, \dots, v_n) \\ f(cv_1, v_2, \dots, v_n) = c f(v_1, v_2, \dots, v_n) \end{cases}$

(ii) alternating if $f(\dots, v_i, v_i, \dots) = 0$ i.e. f vanishes if two adjacent slots are equal

Rmk: { (i) + (ii) $\Rightarrow$ $f(\dots, v_i, w_i, \dots) = -f(\dots, w_i, v_i, \dots)$ Pf. Consider $f(\dots, v+w, v+w, \dots)$
 easy exercises { $f(\dots, v_i, \dots, v_i, \dots) = 0$
 $f(\dots, v_i, \dots, w_i, \dots) = -f(\dots, w_i, \dots, v_i, \dots)$ swapping any two slots changes sign of f

Proof of thm Step 1 (There is at most one such $f^n D$). Let D be such a f^n .

$$A = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \dots & a_{nn} \end{bmatrix} = [C_1, \dots, C_n], \text{ where } C_j = \begin{bmatrix} a_{1j} \\ \vdots \\ a_{nj} \end{bmatrix} = a_{1j} e_1 + \dots + a_{nj} e_n$$

$$D(A) \xrightarrow[\text{multilinearity}]{\text{using}} \sum_{\sigma \text{ set of all functions } \{1, \dots, n\} \rightarrow \{1, \dots, n\}} a_{\sigma(1)1} \dots a_{\sigma(n)n} D(e_{\sigma(1)}, \dots, e_{\sigma(n)}) \quad \text{sum of } n^n \text{ terms}$$

This finishes step 1.

Step 2
For existence
will use $l(\sigma)$
 $\text{sgn}(\sigma) = (-1)^{l(\sigma)}$
where $l(\sigma)$ is
of inversions

using
alternating
nature

$\{1, \dots, n\} \rightarrow \{1, \dots, n\}$

$$\sum_{\sigma \text{ set of all permutations of } \{1, \dots, n\}} a_{\sigma(1)1} \dots a_{\sigma(n)n} \underbrace{D(e_{\sigma(1)}, \dots, e_{\sigma(n)})}_{\pm 1} \quad \begin{array}{l} \text{sum of } n! \text{ terms} \\ \text{so } \{\sigma(1), \dots, \sigma(n)\} = \{1, \dots, n\} \\ \text{sign depending on no. of swaps needed to get } (e_1, \dots, e_n) \end{array}$$